

Bryce A. Stelle
stelleba@rose-hulman.edu
(317) 823-8570
11033 Sanabria Drive - Indianapolis, Indiana 46235

Objective: Computer Science undergraduate seeking a full-time software engineering position beginning in 2027. Eager to apply programming skills and problem solving capabilities to help improve the lives of others.

Education: **Bachelor of Science, Computer Science** **May 2027 | GPA: 3.88/4.0**
Rose-Hulman Institute of Technology, Terre Haute, IN
Related Courses: Object-Oriented Software Development, Intro to Web Programming, Data Structures & Algorithm Analysis, Computer Architecture, Programming Language Concepts, Operating Systems, Database Systems, Intro to Parallel Computing, Software Requirements Engineering, Technical Communications, Theory of Computation

Skills:

Software: Windows, Linux

Programming: Python, Java, Qt Modeling Language (QML), C++, C#, C, JavaScript
HTML, CSS, Assembly, Verilog, Racket/Scheme

Technical: Computer Aided Design, Bench Saw, Laser Cutter, Lathe, Band Saw,
Soldering, Plastic welding

Other: Digital Art & Graphics, Scrum methodology, Presentation &
Public Speaking

Experience: **Viavi Solutions** **May 2026 - Aug. 2026**
Software Engineering Intern **May 2025 - Aug. 2025**
May 2024 - Aug. 2024

- Wrote embedded software to run on a fiber-optic testing device, making use of SSH and VNC.
- Used Claude AI often to significantly boost productivity and programming capabilities, reducing multiple days of work.
- Tracked down and fixed (various) bugs on proprietary products through learning how to navigate a large-scale codebase.
- Learned how to read and debug C++ files by learning C++ class structure and notations, as well as the libraries and frameworks used by the company.
- Presented implementations (features, bug fixes, etc.) in front of marketing to ensure the software was working as expected.
- Integrated Global Navigation Satellite System (GNSS) features into the device by adding storage for data, a responsive icon to tell the user when the device has a 3-dimensional lock on position, and a settings configuration page.
- Fixed delay between releasing of a network resource and being able to use it by modifying how often Linux's NetworkManager renews its IP addresses / Domain Name System (DNS) servers.
- Learned how to write JenkinsFile and Bamboo YAML files for software builds, as well as run and debug them.
- Implemented and improved testing pipeline for embedded devices.

Projects: **Personal Projects**

- **Nachhall (Frieren Fan Game) - Ongoing**
Sole Developer
 - Implemented Jump-Flood Algorithm (JFA) to recreate 2d screenspace VFX, generating specialized outlines and distance effects.
 - Learned and implemented procedural generation

- Implemented lighting-colored outlines via my own method.
- Optimized for lower-end computers.
- Implemented a chord (multi-key) input system for skill input bindings.

Rose-Hulman Institute of Technology

- **Senior Capstone Project - Encrypted Medication Record**

Developer

- In-progress development of wearable, encrypted device to enable quick ease of access to a patient's medication records in the case of an emergency.

- **Parallel Polygon Rendering**

Sole Developer and Tester

- Implemented 3D polygon rendering with occlusion-culling, clipping, and transparency.
- Translation of the input into objects and the graphics processing was handled in parallel, utilizing the same techniques as a GPU.

- **Course Planner**

Primary Developer and Tester

- Improved pre-existing major-specific flowcharts by creating an interactive map of a student's 4-year career that follows the same format.
- Designed for usability far into the future by adding flexibility with custom courses.
- User-data storage implemented via asynchronous programming, promises & calls to the database.

- **Auto-balancing Adelson-Velsky and Landis (AVL) Tree**

Co-Developer and Tester

- Implemented a height-balanced tree with rotations, addition and deletion.
- Each method was ensured to be efficient using big-O runtime analysis.

Awards:	Alpha Lambda Delta Honor Society Member	Dec. 2023 - Present
	Rose-Hulman Merit Scholarship	Sep. 2023 - Present
	Dean's List	Sep. 2023 - Present
	Taekwondo Black Belt	Aug. 2012

Activities:	English (ESL) Tutor , Rose-Hulman Institute of Technology	Aug. 2025 - Present
	Helped tutor students whose primary language isn't English, participated in ESL events with other tutors and other students.	

Pickleball Club , Rose-Hulman Institute of Technology	Nov. 2024 - Present
President & Communications Officer: Organized meetings, reserved spaces, produced flyers and announced club events.	

Martial Arts Club , Rose-Hulman Institute of Technology	Jan. 2024 - Oct. 2024
Trained in MMA with techniques from Muay Thai and Boxing.	
